

-c is used for creating a new snapshot. To revert back to a particular state, use the following command:

```
$qemu-img snapshot -a 5 test.img
```

...where 5 is the snapshot ID. To see all the available snapshots, use the command given below:

```
$qemu-img -l test.img
```

To delete a snapshot, use the command shown below:

```
$qemu-img snapshot -d 2 /images/sles11sp1.qcow2
```

Securing your virtual storage: The other advantage of using file storage is that it's easy to ship like a container. But with shipping comes the responsibility of securing your storage. So a good solution for securing your virtual data storage is to protect it by using encryption, for which you will need a password every time to mount it. For encrypting your storage, let's use *dm-crypt*. We will try encryption on fresh file storage, as follows:

```
$truncate encrypted.raw --size=2GB
```

Next, set up a LUKS header, as follows:

```
$cryptsetup luksFormat encrypted.raw
```

Warning:

Don't try this with an already formatted partition because it will delete all the previous data inside the partition.

This will prompt you to enter a fresh password. To gain access to the device, use the command given below:

```
$cryptsetup open encrypted.raw my_encp.raw
```

my_encp.raw is the name of the file where our partition is mapped in */dev/mapper/*. Now you can create a file system on top of it, as follows:

```
$mkfs.fstype /dev/mapper/my_encp.raw
```

Mount the newly created partition anywhere with the following command:

```
$mount -t ext2 /dev/mapper/my_encp.raw /mnt/
```

Once use of the storage is finished, you can unmount it, as follows:

```
# umount /mnt/
```

```
# cryptsetup close my_encp.raw
```



Note: You can do the same disk scaling and other operations on disk, but now you need to make changes to */dev/mapper/my_encp.raw* rather than */dev/loop0*.

Using a file storage virtual machine: A great benefit of file storage is its use as a base storage for virtual machines. Once you decide to use a file storage for an OS run on a virtual machine, other operations like scaling and encryption can also be applied to it. To create a virtual machine instance from our already created storage, let's use the *qemu-kvm* from our already created storage, as shown below:

```
$qemu-kvm -name "my_os" -m 1024 -smp 2 -drive file=test.
img,if=virtio,\
index=0,media=disk,format=raw -drive file=ubuntu-14.04.
iso,index=1,media=cdrom
```

This will start a new virtual machine with the minimal option selected. Here, *-m* defines the amount of RAM allocated and *-smp* refers to the number of cores. You can read about more available options in the *qemu-kvm* man page or by using *qemu-kvm --help*. To start an already created virtual machine, use the following command:

```
$qemu-kvm -name "my_os" -m 1024 -smp 2 -drive\ file=/images/
sles11/hda,if=virtio,index=0,media=disk,format=raw
```

File storage can be a good solution for your enterprise environment or personal use, but what matters is how you make it useful at a different level of your cloud solution. Storage management in the case of file storage doesn't end here since there is a lot more you can do with it. It can be a powerful as well as flexible solution for storage management, but requires thorough knowledge to get the job done. 

References

- [1] For lsyncd: <http://www.linuxtechin.com/install-and-use-lysyncd-on-centos-7-rhel-7/>
- [2] For qemu utility: https://www.suse.com/documentation/sles11/book_kvm/data/book_kvm.html
- [3] For Dm-crypt encryption: https://wiki.archlinux.org/index.php/Dm-crypt/Encrypting_a_non-root_file_system

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